



01 — Ingestion & Episodic Encoding

Overview

The ingestion pipeline converts raw text into an episodic memory trace: a dense vector embedding stored in FAISS + SQLite, with an initial salience derived from its novelty relative to existing memories.

Mathematical Formulation

Step 1: Text Encoding

Given a text string $t \in \Sigma^*$:

$$\mathbf{e} = \text{encode}(t) \in \mathbb{R}^d, \quad d = 384$$

Where `encode` is the ONNX Runtime inference of `paraphrase-multilingual-MiniLM-L12-v2`, producing an L2-normalized vector:

$$\|\mathbf{e}\|_2 = 1$$

Step 2: Salience Initialization (Novelty)

For a new episode with embedding $\mathbf{e}$, find its nearest neighbor in the episodic store:

$$\text{nn_sim} = \max_{i \in \mathcal{E}} \cos(\mathbf{e}, \mathbf{e}_i) = \max_{i \in \mathcal{E}} \langle \mathbf{e}, \mathbf{e}_i \rangle$$

The initial salience is the complement of similarity (more novel $\rightarrow$ higher salience):

$$s_0 = \max(s_{\min}, w_{\text{novelty}} \cdot \frac{1 - \text{nn_sim}}{2})$$

Where:

- $s_{\min} = \text{min_salience}$ (default: `0.01`)
- $w_{\text{novelty}} = \text{novelty_weight}$ (default: `1.0`)

Step 3: Storage

The episodic memory m is stored as:

$$m = (\text{id}, \text{event_id}, \text{ts}, \mathbf{e}, s_0, \text{metadata})$$

- **FAISS Index:** `IndexIDMap2(IndexFlatIP(d))` — inner product on normalized vectors = cosine similarity.
- **SQLite:** Row in `episodic_memories` table with salience, timestamp, metadata_json.

Configuration

EpisodicStoreConfig

Parameter	Type	Default	Description
<code>dim</code>	<code>int</code>	<code>384</code>	Embedding dimension
<code>db_path</code>	<code>str</code>	<code>"slowwave.db"</code>	SQLite database path
<code>faiss_index_path</code>	<code>str</code>	<code>"episodic.faiss"</code>	FAISS index persistence path

SalienceConfig (initialization)

Parameter	Type	Default	Description
<code>novelty_weight</code>	<code>float</code>	<code>1.0</code>	Weight for novelty component
<code>min_salience</code>	<code>float</code>	<code>0.01</code>	Floor for salience values

EncoderConfig

Parameter	Type	Default
<code>model_name</code>	<code>str</code>	<code>"sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2"</code>
<code>normalize</code>	<code>bool</code>	<code>True</code>

Parameter	Type	Default	
<code>device</code>	<code>str</code>	<code>"cpu"</code>	
<code>use_onnx</code>	<code>bool</code>	<code>True</code>	
<code>cache_dir</code>	<code>`str`</code>	<code>None`</code>	

Key Invariants

1. All embeddings are L2-normalized; cosine similarity reduces to dot product.
2. `min_salience` ensures no trace decays to exactly zero.
3. `novelty_weight = 1.0` means a perfectly novel episode (`nn_sim = -1`) gets salience = 1.0; an identical duplicate (`nn_sim = 1`) gets salience = `min_salience`.
4. FAISS `IndexFlatIP` + `IndexIDMap2` means exact search, not approximate — full precision at the cost of linear scan.